M4010104 / M4020104: PROGRAMMING IN PYTHON · DIGITAL UNIVERSITY KERALA · CAPSTONE PROJECT SPECIFICATION

Project 14: Utility Solar PV Inverter DC-to-AC Conversion Efficiency & Soiling Detector

Effort Budget: 10–12 Hours | Core Stack: Pure Python, NumPy, Pandas, Matplotlib, Seaborn, Plotly | Max Marks: 25 | Instructor: Dr. Sunil TT

1. Executive Overview & Domain Context

Utility-scale grid-connected photovoltaic (PV) power plants monitor 15-minute generation metrics across individual solar inverters alongside plant-level meteorological sensors measuring ambient temperature, module surface temperature, and solar irradiation (W/m2). Equipment degradation, inverter tripping, and dust soiling directly erode power yield.

Your team will build an industrial diagnostic tool in pure Python that parses synchronized inverter generation and weather sensor tables, cleans negative power readings and night-time noise, computes DC-to-AC conversion efficiency (η = P_AC / P_DC) in NumPy, models thermal derating degradation above 45°C, and detects underperforming units answering the 12 core questions below.

Primary Public Dataset & Telemetry/Data Schema

The project utilizes the verified public Indian Solar Power Generation & Sensor Data on Kaggle (**`https://www.kaggle.com/datasets/anikannal/solar-power-generation-data`**). The dataset structure captures:

- **Raw Feature Attributes:** **`DATE_TIME`**, **`PLANT_ID`**, **`SOURCE_KEY`** (inverter ID), **`DC_POWER`** (kW), **`AC_POWER`** (kW), **`DAILY_YIELD`**, **`AMBIENT_TEMPERATURE`**, **`MODULE_TEMPERATURE`**, **`IRRADIATION`**.
- **Engineered / Target Variables:** **`conversion_efficiency`** (AC / DC), **`expected_power`**, **`power_residual`**, **`soiling_loss_factor`**, **`fault_flag`**.

2. Complete Project Directory & File Architecture

Each team must organize their repository into a standardized, production-grade layout. All application source code must reside inside a dedicated package (**`src/`**), with separate directories for raw/curated data, interactive development notebooks, visual figures, and formal documentation:

```
solar-inverter-efficiency-diagnostics/
├── README.md                          # Repository documentation & GitHub Pages source
├── requirements.txt                   # Dependency specification (numpy, pandas, matplotlib, seaborn, p
├── run_pipeline.py                    # Standalone production CLI application entry point
├── data/
│   ├── raw/
│   │   └── solar_power_generation_raw.csv            # Raw uncleaned input dataset (including simulated
│   └── processed/
│       ├── clean_solar_power.csv          # Curated, validated clean CSV dataset
│       └── solar_inverter_audit.log           # Timestamped audit report detailing filtered/rejected re
├── src/
│   ├── __init__.py                    # Python package initialization
│   ├── ingestion.py                   # Defensive line-by-line file reader, range auditor & CSV exporte
│   ├── analyzer.py                    # SolarInverterDiagnostics OOP class, NumPy vectorization & Panda
│   └── visualizer.py                  # Matplotlib envelope plots, Seaborn heatmaps & Plotly HTML expor
├── notebooks/
│   └── development_cycle.ipynb        # Development cycle notebook: EDA, trials, tests & unit assertion
└── reports/
    ├── figures/
    │   ├── primary_envelope.png       # Matplotlib confidence envelope & regression overlay
    │   ├── correlation_heatmap.png    # Seaborn annotated feature correlation heatmap
    │   └── distribution_regimes.png   # Seaborn distribution across categorical/domain tiers
```

```
├── interactive_dashboard.html     # Standalone interactive Plotly HTML dashboard with controls
└── Project_Report.pdf             # 3-4 page final technical report
```

3. Dual Development Deliverables: Notebook vs. Standalone Program

To mirror industry software engineering standards, each student group must deliver two distinct, complementary implementations of their solution:

Deliverable A: The Development Cycle Notebook (`notebooks/development_cycle.ipynb`)

The development notebook chronicles your team's engineering journey from raw data exploration to validated model prototypes. It is not merely a final script pasted into cells, but an authentic record of the development cycle:

- **Phase 1: Exploratory Data Ingestion & Diagnostics:** Loading raw data, inspecting missing values, testing boundary rules, and logging rejected records.
- **Phase 2: Mathematical Formulation & Feature Verification:** Developing and validating NumPy array calculations against target benchmarks using `np.allclose()`.
- **Phase 3: OOP Class Architecture & Unit Tests:** Instantiating the `SolarInverterDiagnostics` class, executing automated unit assertions (e.g., verifying that passing invalid parameters raises an explicit `ValueError`), and testing edge cases.
- **Phase 4: Statistical Modeling & Regimen Exploration:** Implementing ordinary least-squares regression, rolling-window calculations, and multi-metric Pandas grouping.
- **Phase 5: Visual Prototype Development:** Iteratively designing and refining Matplotlib and Seaborn figures, followed by configuring the interactive Plotly HTML dashboard.

Deliverable B: The Final Standalone Python Program (`run_pipeline.py` & `src/`)

While the notebook documents the exploratory development cycle, your repository must also provide a production-ready, fully executable Python application. The program must run end-to-end directly from the terminal without requiring Jupyter:

- **Command-Line Interface (CLI):** Executed via `python run_pipeline.py --input data/raw/solar_power_generation_raw.csv --output-dir data/processed --report-dir reports`.
- **Modular Package Import:** The driver script imports cleanly separated domain logic from `src.ingestion`, `src.analyzer`, and `src.visualizer`.
- **Terminal Feedback & Exit Codes:** The program must print structured progress indicators, log clean/faulty record counts to the console, and return an exit code of `0` upon successful pipeline completion.
- **Automated Artifact Generation:** In a single execution pass, the script reads raw inputs, writes `data/processed/clean_solar_power.csv`, saves `data/processed/solar_inverter_audit.log`, exports static figures to `reports/figures/`, and generates the standalone `reports/interactive_dashboard.html`.

4. Technical Investigation Requirements & Research Questions (Q1–Q12)

Your team's implementation across the development notebook and standalone program must rigorously answer the following 12 course-aligned engineering questions:

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
File I/O & Exceptions (`src/ingestion.py`)	**Q1**	**Inverter Timestamp & Negative Power Audit:** Parse raw generation CSV rows using try-except. Validate datetime strings, filter out night-time negative power noise (DC_POWER < 0 when Irradiation == 0), and log faulty rows.	open(), try-except (ValueError, IndexError), datetime parsing

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Data Curation & Export (`src/ingestion.py`)	**Q2**	**Curated Solar Generation Export & Audit Log:** Write clean daylight records to clean_solar_power.csv while logging inverters with missing intervals or continuous zero-output into solar_inverter_audit.log.	Defensive file writing, dictionary parsing, formatted log reporting
OOP Design (`src/ingestion.py`)	**Q3**	**Solar Inverter Diagnostics Class:** Design an encapsulated class SolarInverterDiagnostics with methods load_from_csv(), detect_inverter_faults(min_efficiency=0.88), and summary_report(). Ensure min_efficiency <= 0 raises ValueError.	OOP class, __init__, encapsulation, raise ValueError
NumPy Arrays (`src/analyzer.py`)	**Q4**	**Vectorized Conversion Efficiency Verification:** Using only raw NumPy arrays of DC and AC power, compute inverter conversion efficiency (η = P_AC / P_DC). Verify daylight averages against benchmarks via np.allclose().	np.divide(), np.where(), np.allclose()
NumPy Masking (`src/analyzer.py`)	**Q5**	**Thermal Derating vs. Inverter Fault Masking:** Flag timestamps where module temperature exceeds 45°C in NumPy. Distinguish temperature-induced efficiency drops from sudden inverter tripping faults using boolean masks.	np.mean(), np.std(), boolean masks (&, |), Python set
NumPy Regression (`src/analyzer.py`)	**Q6**	**Irradiation vs. DC Power Ordinary Least-Squares Fit:** Fit an ordinary least-squares model (DC_Power = b0 + b1*Irradiation + b2*Module_Temp) via np.linalg.lstsq. Quantify the thermal derating coefficient (negative slope of temperature).	np.column_stack(), np.linalg.lstsq(), matrix product (@), R2
Pandas Series (`src/analyzer.py`)	**Q7**	**Inverter Yield Summary & Missing Data Audit:** Quantify total daily yield and missing 15-minute intervals per inverter in Pandas. Identify underperforming inverters consistently generating below plant median.	.groupby('SOURCE_KEY').agg(), .isna().sum()
Pandas Groupby (`src/analyzer.py`)	**Q8**	**Irradiation-by-Temperature Grid & Pivot Table:** Discretize Irradiation into Low, Moderate, and Peak; discretize Module_Temp into Cool, Warm, and Hot. Build a 3×3 pivot_table(margins=True) showing mean AC power.	pd.cut(), .groupby().agg(), .pivot_table(margins=True)
Pandas Reshaping (`src/analyzer.py`)	**Q9**	**Wide-to-Long Inverter Output Ranking via Melt:** Reshape daily AC power generation across 22 inverters into tidy long format via pd.melt() and extract top 5 consistently highest-performing inverters.	pd.melt(), .groupby('SOURCE_KEY'), .nlargest(5)
Matplotlib (`src/visualizer.py`)	**Q10**	**Matplotlib Diurnal Generation & Thermal Envelope:** Plot diurnal DC and AC power curves over a sample 24-hour cycle in Matplotlib. Shade the irradiation envelope on a secondary y-axis with complete legends.	plt.subplots(), ax.plot(), ax.twinx(), ax.fill_between()
Seaborn (`src/visualizer.py`)	**Q11**	**Seaborn Inverter Efficiency Boxplots & Heatmap:** Render side-by-side Seaborn boxplots comparing conversion efficiency across all 22 inverters. Generate a module-temp vs. efficiency jointplot with sns.jointplot(kind='reg').	sns.boxplot(), sns.jointplot(), sns.despine()

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Plotly HTML (`src/visualizer.py`)	Q12	**Interactive Solar Plant Diagnostics Dashboard:** Build an interactive HTML dashboard (interactive_dashboard.html) displaying plant generation with inverter multi-select filters, efficiency heatmaps, and time sliders.	plotly.express.line(), custom hover tooltips, write_html()

5. Collaborative Git Workflow, GitHub Pages & README.md Specification

Best Practices: Collaborative Git Workflow & GitHub Pages Deployment from Day 1

From the very first day, the student team must adopt professional version control and documentation practices. On Day 1, one designated team member initializes the central public GitHub repository, establishes a clear branch protection rule preventing direct pushes to `main`, adds all teammates as registered **Contributors** (with write access), and grants read access to the Course Instructor (`Dr. Sunil TT`) and teaching assistants. Immediately thereafter, the team must activate **GitHub Pages** (via *Settings → Pages → Build and deployment: Deploy from a branch: main /docs or root*) so that project documentation, interactive architecture diagrams, and the live Plotly dashboard (`reports/interactive_dashboard.html`) are publicly accessible and continuously deployed throughout the development lifecycle. All feature development must follow a strict branching model (e.g., `feature/ingestion`, `feature/analysis`, `feature/visualization`): individual team members must write small, atomic, well-commented commits from their own personal GitHub accounts and merge them into `main` exclusively through formal **Pull Requests (PRs)** accompanied by peer review. Dumping entire codebases in a single, monolithic end-of-semester commit is strictly prohibited and will result in zero marks for individual Git contribution; maintaining an unbroken, transparent Git commit and PR audit trail is essential for passing the code ownership defense during the live viva-voce examination.

Repository README.md Specification Template

The root `README.md` serves as the primary technical landing page and must strictly conform to the following structural format:

````
# Project 14: Utility Solar PV Inverter DC-to-AC Conversion Efficiency & Soiling Detector
[![Course](https://img.shields.io/badge/Course-Programming%20in%20Python-blue.svg)](https://duk.ac.in)
[![Python](https://img.shields.io/badge/Python-3.11%2B-brightgreen.svg)](https://python.org)
[![License](https://img.shields.io/badge/License-MIT-yellow.svg)](LICENSE)

## 1. Project Overview & Domain Context
- **Institution:** Digital University Kerala (MTech AI / MSc AI)
- **Course:** M4010104 / M4020104 – Programming in Python | **Instructor:** Dr. Sunil TT
- **Dataset:** Indian Solar Power Generation & Sensor Data on Kaggle (https://www.kaggle.com/datasets/an

## 2. System Architecture & Pipeline Design
[Provide an ASCII or Mermaid architecture flow diagram showing raw ingestion -> validation -> analyzer -

## 3. Installation & Environment Setup
```bash
git clone https://github.com/<your-team>/solar-inverter-efficiency-diagnostics.git
cd solar-inverter-efficiency-diagnostics
python -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate
pip install -r requirements.txt
```

## 4. Execution Instructions
### A. Running the Standalone Application
```bash
python run_pipeline.py --input data/raw/solar_power_generation_raw.csv --output-dir data/processed --rep
```
### B. Running the Development Notebook
```bash
````
````

```
jupyter notebook notebooks/development_cycle.ipynb
```

5. Key Analytical & Statistical Findings (Q1-Q12 Summary)
[Provide a concise markdown summary table detailing raw vs. clean counts, regression coefficients, anoma

6. Interactive Telemetry Dashboard
Explore the live interactive Plotly visualization hosted on GitHub Pages:
🔗 [Live Project Dashboard](https://<your-team>.github.io/solar-inverter-efficiency-diagnostics/interact

7. Contributor Git Commit & Pull Request Log Table
Contributor Name	GitHub Handle	Assigned Module / Tasks	Verified Commit Hash Range	Merged Pull
[Student Full Name]	`@github_handle`	Modular Architecture (`src/ingestion.py`, `src/analyzer.py`,

6. Evaluation Rubric & Mandatory Academic Integrity Governance

Projects are evaluated on a 25-mark scale based on code quality, mathematical correctness, visualization depth, git collaboration discipline, and live technical defense:

Assessment Criterion	Basis of Evaluation & Deliverable Verification	Max Marks
1. Data Curation & File I/O Pipeline (Q1–Q2)	Robust line-by-line file reading with defensive **`try-except`** handling, physical/domain bounds validation, clean CSV output, and structured **`solar_inverter_audit.log`** reporting.	**5 Marks**
2. OOP, NumPy & Pandas Analytics (Q3–Q9)	Modular **`SolarInverterDiagnostics`** class architecture, vectorized NumPy feature re-derivations (verified via **`np.allclose`**), regression modeling, and Pandas rolling/groupby/melt operations.	**6 Marks**
3. Multi-Library Visualizations (Q10–Q12)	Publication-quality Matplotlib confidence envelope/trend plot, Seaborn correlation heatmap and distribution violin plot, and standalone interactive Plotly HTML dashboard.	**5 Marks**
4. Software Packaging, GitHub & Documentation	Clean directory layout, comprehensive **`README.md`**, working GitHub Pages site, **`development_cycle.ipynb`** notebook, standalone **`run_pipeline.py`**, and balanced individual Git commits/PRs.	**4 Marks**
5. Live Presentation & Code Viva-Voce Defense	Technical walkthrough and individual live viva defense where each student must explain or modify lines of code in real time without external assistance.	**5 Marks**

> **STRICT ACADEMIC INTEGRITY & ZERO-TOLERANCE AI POLICY (0 MARKS PENALTY):** All submitted Python source code (**`.py`**, **`.ipynb`**) and written documentation will be screened through an automated **AI-Code & Plagiarism Detection Suite**. If AI-generated code or synthetic documentation is detected at any checkpoint or final submission, **0 Marks will be awarded immediately** for the entire project. Each student must prove authentic individual code ownership during the live viva-voce examination — *Course Instructor: Dr. Sunil TT*.

7. Milestone Schedule, Submission Checklist & Viva Defense Prompts

To ensure systematic progress across the semester, teams must adhere to three project checkpoints and prepare for on-the-spot live code modification challenges during the viva-voce examination:

Milestone Stage	Scheduled Window	Required Deliverables & Verification Milestones
Checkpoint 1 (Data Ingestion)	Week 11 Lab Session	Repository initialized with contributors and GitHub Pages active. **`src/ingestion.py`** operational with line-by-line filtering, **`clean_solar_power.csv`** exported, and **`solar_inverter_audit.log`** generated.
Checkpoint 2 (Analytics & OOP)	Week 13 Lab Session	**`src/analyzer.py`** complete with **`SolarInverterDiagnostics`** class, NumPy feature checks (Q4), outlier masks (Q5), regression modeling (Q6), and exploratory notebook sections.
Final Submission & Viva	Week 15 Project Review	Full package operational: standalone **`run_pipeline.py`**, polished **`development_cycle.ipynb`**, interactive **`interactive_dashboard.html`** on GitHub Pages, 3–4 page PDF report, and live code defense.

Representative Live Viva-Voce Code Modification Challenge

During the individual code viva-voce, the instructor will ask students to execute an immediate 2-minute code modification on their working repository to demonstrate genuine authorship:

- **Challenge Prompt:** *"In src/analyzer.py, filter for peak solar hours (Irradiation > 0.6 kW/m2) and identify inverters with conversion efficiency < 0.90. Re-run your module and display the underperforming inverter IDs."*